

INSTITUTE OF ACTUARIES OF INDIA

Subject SA1 - Health and Care

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- a) A chronic illness is a long-term health condition that typically lasts for an extended period, often for years or even a lifetime. These conditions usually do not have a cure, but they can often be managed with medication, therapy, lifestyle changes, or other treatments. Chronic illnesses can vary widely in their severity, impact on daily life, and prognosis. Diabetes, heart disease, arthritis, asthma, and some mental health issues like bipolar disorder or depression are a few examples of chronic illnesses. Managing a chronic illness often requires ongoing medical care and attention to maintain quality of life and minimize symptoms.
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- b) Analyzing the impact of chronic diseases in the insurance sector is crucial for several reasons:
- 1) **Risk Assessment:** Chronic diseases significantly impact an individual's health and longevity, which affects their insurability. Insurance companies need to assess the risk associated with covering individuals with chronic illnesses accurately. Analyzing the prevalence, severity, and progression of chronic diseases helps insurers determine appropriate premiums and coverage limits.
 - 2) **Underwriting:** Understanding the prevalence and impact of chronic diseases enables insurers to develop effective underwriting criteria. By analyzing data related to chronic illnesses, insurers can assess the likelihood of claims and tailor policies accordingly. This ensures fair pricing and risk management for both insurers and policyholders.
 - 3) **Product Development:** Insights into the impact of chronic diseases inform the development of insurance products tailored to the needs of individuals with these conditions. For example, insurers may offer specialized health insurance plans or riders that cover specific chronic illnesses, providing financial protection for medical expenses related to managing these conditions.
 - 4) **Claims Management:** Chronic diseases often require ongoing medical treatment, which can result in frequent claims for insurers. By analyzing the impact of chronic diseases, insurers can anticipate claim patterns, allocate resources effectively, and streamline claims processing to provide timely assistance to policyholders.
 - 5) **Healthcare Costs:** Chronic diseases contribute significantly to healthcare costs, including medical treatment, medication, and related services. Insurers need to understand these costs to set premiums that reflect the expected expenses associated with covering individuals with chronic illnesses. Analyzing the impact of chronic diseases helps insurers manage healthcare costs and ensure the long-term sustainability of insurance programs.
 - 6) **Customer Engagement and Support:** Insurance companies can use insights into chronic diseases to provide proactive support and resources to policyholders managing these conditions. This may include wellness programs, disease management initiatives, and access to healthcare professionals or support networks. By offering tailored support, insurers can improve customer satisfaction, promote health outcomes, and reduce the likelihood of costly claims.
 - 7) **Regulatory Compliance:** Regulatory requirements often mandate that insurers analyze and report on the impact of chronic diseases as part of their risk management and financial planning processes. Compliance with these regulations ensures transparency, accountability, and stability within the insurance sector.

- 8) Actuarial Modeling: Analyzing the impact of chronic diseases helps actuaries develop accurate models for predicting future claims and liabilities, which is essential for setting reserves and maintaining solvency.
- 9) Reinsurance Strategies: Understanding the prevalence and severity of chronic diseases allows insurers to make informed decisions about reinsurance arrangements to mitigate their exposure to high-risk populations or conditions.
- 10) Investment Strategies: Chronic diseases can influence investment strategies for insurance companies, as they may need to allocate resources to cover future claims or invest in healthcare-related assets.
- 11) Market Segmentation: Insight into the impact of chronic diseases enables insurers to segment the market effectively and target specific demographic groups or individuals with tailored insurance products.
- 12) Data Analytics: Advanced data analytics techniques can be applied to analyze large datasets related to chronic diseases, providing valuable insights for risk management, customer profiling, and predictive modeling.
- 13) Fraud Detection: Analyzing patterns related to chronic diseases can help insurers identify potential fraud or abuse within their claims processes, improving the integrity of insurance operations.
- 14) Policyholder Education: Insurance companies can use insights into chronic diseases to educate policyholders about risk factors, prevention strategies, and available resources for managing their health effectively.
- 15) Provider Partnerships: Collaboration with healthcare providers and institutions can enhance insurers' ability to analyze and address the impact of chronic diseases, fostering innovation in care delivery and cost containment efforts.
- 16) Government Relations: Analysis of chronic disease data can inform insurers' advocacy efforts with government agencies and policymakers, influencing healthcare policy development and regulatory reforms.
- 17) Long-Term Care Planning: Chronic diseases often necessitate long-term care, and insurers may offer products or services tailored to this need, such as long-term care insurance or annuities with healthcare benefits.
- 18) Global Health Trends: Monitoring global trends in chronic diseases allows insurers to adapt their strategies to evolving health challenges, including emerging diseases and shifting demographic patterns.
- 19) Employer Benefits: Employers often rely on insurance benefits to attract and retain talent, and understanding the impact of chronic diseases helps insurers design employer-sponsored health plans that meet workforce needs.
- 20) Customer Retention: By providing comprehensive coverage and support for chronic diseases, insurers can enhance customer loyalty and retention, reducing churn and maximizing lifetime customer value.
- 21) Public Health Initiatives: Insurance companies can contribute to public health initiatives by leveraging their resources and expertise to promote awareness, prevention, and early intervention for chronic diseases.

- 22) Health Equity: Analysis of chronic disease data can highlight disparities in healthcare access and outcomes, prompting insurers to address inequities and promote health equity initiatives.
- 23) Technological Innovation: Advancements in technology, such as wearable devices and telemedicine platforms, can facilitate remote monitoring and management of chronic diseases, offering new opportunities for insurers to improve care and reduce costs.
- 24) Patient-Centered Care: Understanding the impact of chronic diseases from the patient's perspective allows insurers to prioritize patient-centered care approaches, focusing on individual needs, preferences, and goals to improve overall satisfaction and health outcomes.
- 25) Preventive Health Initiatives: Analysis of chronic disease data can inform insurers' investments in preventive health initiatives, such as screening programs, vaccination campaigns, and wellness incentives, which can help reduce the incidence and severity of chronic conditions over time.

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- c) The report should equip the board of directors with valuable insights and recommendations to make informed decisions about addressing the challenges posed by chronic diseases within the insurance industry. Following is a comprehensive outline for a report on analyzing the impact of chronic diseases on insurance policies to be presented to the board of directors of an insurance company.

1. Executive Summary:

- a) Brief overview of the report's purpose, findings, and recommendations
- b) Highlights key points regarding the impact of chronic diseases on insurance policies

2. Introduction:

- a) Provide background information on the insurance company, including its history, mission, and target market.
- b) Explain the importance of analyzing chronic diseases for insurance policies, highlighting the significant impact these conditions have on individuals' health and healthcare costs.
- c) Clearly state the objectives of the report, such as identifying challenges, assessing current policies, and proposing recommendations for improvement.

3. Overview of Chronic Diseases:

- a) Define chronic diseases and provide examples to illustrate their diverse nature.
- b) Present statistics on the prevalence of chronic diseases globally and within the company's customer base.
- c) Discuss the multifaceted impact of chronic diseases, including their effects on individuals' quality of life, healthcare utilization, and economic burden.
- d) Highlight key trends and projections related to chronic diseases, such as increasing prevalence rates due to aging populations and changes in lifestyle factors.

4. Analysis of Current Insurance Policies:

- a) Review the similar insurance products currently offered by the company, focusing on coverage options for chronic diseases.
- b) Evaluate the underwriting criteria used in the similar products to assess applicants with chronic conditions and determine their insurability.

- c) Discuss the company's premium pricing strategies for policies covering chronic diseases, considering factors such as risk assessment and market competitiveness.
 - d) Examine the claims management processes in place for handling claims related to chronic diseases, including reimbursement procedures and customer support.
- 5. Data Analysis and Insights:
 - a) Analyze internal data collected by the company, such as policyholder demographics, claims data, and healthcare utilization patterns.
 - b) Incorporate external data sources, such as epidemiological studies and healthcare industry reports, to provide context and validate findings.
 - c) Identify trends and insights gleaned from the data analysis, such as high claim frequency for certain chronic conditions or regional variations in disease prevalence.
 - d) Use statistical techniques to quantify the financial impact of chronic diseases on the company's operations, such as loss ratios and reserve adequacy.
- 6. Market Research and Competitive Landscape:
 - a) Conduct market research to assess competitors' approaches to covering chronic diseases, including product offerings, pricing strategies, and customer service initiatives.
 - b) Identify emerging trends and best practices in the insurance industry related to chronic disease management.
 - c) Benchmark the company's performance against industry standards and key competitors, highlighting areas of strength and opportunities for improvement.
- 7. Regulatory Considerations:
 - a) Review relevant regulations governing insurance policies and chronic diseases, such as anti-discrimination laws and mandates for coverage of essential health benefits.
 - b) Assess the company's compliance with existing regulations and identify potential areas of risk or non-compliance.
 - c) Anticipate regulatory changes that may impact insurance policies covering chronic diseases and evaluate the potential implications for the company's operations.
- 8. Financial Implications:
 - a) Evaluate the financial performance of the company in relation to chronic diseases, considering metrics such as underwriting profitability, claims costs, and overall profitability.
 - b) Assess the adequacy of reserves and risk management strategies for covering potential liabilities associated with chronic diseases.
 - c) Explore opportunities for revenue growth and cost containment initiatives, such as expanding into new markets or implementing efficiency improvements.
- 9. Customer Experience and Satisfaction:
 - a) Gather feedback from policyholders regarding their experiences with insurance policies covering chronic diseases, including satisfaction levels and areas for improvement.
 - b) Identify pain points or gaps in customer service related to chronic disease management and propose solutions to enhance the overall customer experience.
 - c) Consider implementing educational programs or support services to empower policyholders to better manage their chronic conditions and navigate the insurance process effectively.

10. Risk Mitigation Strategies:
 - a) Recommend specific risk mitigation strategies to address the challenges posed by chronic diseases, such as refining underwriting criteria, adjusting pricing models, or enhancing claims management processes.
 - b) Advocate for investments in preventive health initiatives and wellness programs aimed at reducing the incidence and severity of chronic conditions among policyholders.
 - c) Highlight the potential benefits of collaboration with healthcare providers and community organizations to implement holistic approaches to chronic disease management.
 11. Innovation and Future Outlook:
 - a) Explore innovative solutions and technologies for managing chronic diseases within the insurance context, such as telemedicine platforms, remote monitoring devices, or predictive analytics tools.
 - b) Consider opportunities for product innovation and differentiation to meet the evolving needs of policyholders with chronic conditions.
 - c) Provide insights into the long-term outlook for the insurance company in addressing the impact of chronic diseases, considering factors such as demographic trends, technological advancements, and regulatory changes.
 12. Conclusion:
 - a) Summarize the key findings and insights presented in the report, emphasizing the significance of addressing the impact of chronic diseases on insurance policies.
 - b) Reinforce the importance of proactive measures to mitigate risks and seize opportunities in the evolving landscape of chronic disease management.
 - c) Encourage the board of directors to consider the recommendations outlined in the report as actionable steps toward enhancing the company's performance and meeting the needs of policyholders.
 13. Recommendations:
 - a) Provide specific, actionable recommendations for the board of directors to consider, addressing areas such as product development, risk management, customer engagement, and regulatory compliance.
 - b) Prioritize recommendations based on their potential impact and feasibility of implementation.
 - c) Support each recommendation with evidence and analysis presented in earlier sections of the report, demonstrating its rationale and potential benefits for the company.
 14. Appendices:
 - a) Include supplementary information, such as detailed data tables, charts, or graphs referenced in the report.
 - b) Attach supporting documentation, such as regulatory guidelines, market research reports, or customer survey results, to provide additional context and validation for key findings.
 15. References:
 - a) List all sources cited throughout the report, including internal data sources, external research studies, regulatory documents, and industry reports.
- (10)
- d) Analyzing the performance of a new health insurance product is crucial for assessing its effectiveness, profitability, and alignment with the company's strategic objectives. The analysis typically involves several key steps and metrics:**

1. Enrollment and Market Penetration:
 - 1) Measure the rate of enrollment in the new health insurance product compared to initial projections and targets.
 - 2) Analyze market penetration by assessing the percentage of the target market covered by the new product.
 - 3) Evaluate demographic characteristics of enrollees to identify trends and target segments.
2. Revenue and Premium Growth:
 - 1) Track premium revenue generated by the new product over time.
 - 2) Monitor premium growth rates to assess the product's attractiveness to customers and its potential for revenue expansion.
 - 3) Compare premium income against initial projections and industry benchmarks.
3. Claims Experience and Loss Ratios:
 - 1) Analyze claims experience associated with the new product, including claim frequency, severity, and trends.
 - 2) Calculate loss ratios (claims paid divided by premiums earned) to evaluate the product's underwriting profitability.
 - 3) Assess the adequacy of reserves and pricing assumptions based on claims experience.
4. Customer Satisfaction and Retention:
 - 1) Gather feedback from policyholders regarding their satisfaction with the new product's coverage, benefits, and customer service.
 - 2) Monitor policyholder retention rates to gauge customer loyalty and identify areas for improvement.
 - 3) Conduct surveys or focus groups to understand customer preferences and perceptions of value.
5. Risk Pool Composition and Adverse Selection:
 - 1) Analyze the risk pool composition to ensure it aligns with underwriting assumptions and expectations.
 - 2) Assess the presence of adverse selection by comparing the health status and utilization patterns of enrollees to those of the general population.
 - 3) Implement risk mitigation strategies if adverse selection is observed, such as adjusting pricing or marketing tactics.
6. Financial Performance and Profitability:
 - 1) Evaluate the overall financial performance of the new health insurance product, considering factors such as revenue, claims costs, administrative expenses, and profitability.
 - 2) Calculate key financial metrics, such as net income, operating margin, return on investment, and return on equity.
 - 3) Conduct sensitivity analysis to assess the product's resilience to changes in economic conditions or market dynamics.
7. Compliance and Regulatory Considerations:
 - 1) Ensure compliance with applicable regulatory requirements and mandates governing health insurance products.
 - 2) Monitor changes in regulatory landscape and adjust product features or operations accordingly.
 - 3) Conduct audits or reviews to verify adherence to regulatory standards and guidelines.
8. Market Share and Competitive Positioning:

- 1) Assess the new product's market share relative to competitors and industry benchmarks.
 - 2) Monitor competitive pricing, features, and marketing strategies to identify areas of differentiation and competitive advantage.
 - 3) Track market trends and shifts in consumer preferences to inform product development and marketing efforts.
9. Cost-Effectiveness and Efficiency:
- 1) Evaluate the cost-effectiveness of administering the new health insurance product, including administrative expenses, distribution costs, and overheads.
 - 2) Identify opportunities for efficiency improvements, such as streamlining processes, leveraging technology, or outsourcing non-core functions.
 - 3) Conduct cost-benefit analysis to assess the return on investment in marketing, distribution, and operational activities.
10. Long-Term Sustainability and Growth Potential:
- 1) Assess the long-term sustainability of the new health insurance product, considering factors such as market demand, regulatory environment, and demographic trends.
 - 2) Identify opportunities for product diversification, expansion into new markets, or strategic partnerships to fuel growth.
 - 3) Develop strategic plans and initiatives to maximize the product's value proposition and ensure its continued relevance in the marketplace.

By systematically analyzing these key aspects of the new health insurance product's performance, XYZ can make informed decisions regarding product development, marketing strategies, pricing adjustments, and risk management initiatives. This ongoing analysis will enable XYZ to optimize the product's performance, enhance customer satisfaction, and achieve sustainable growth in the competitive healthcare insurance market.

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e)

I. The Insurance Company

Advantages:

- 1) Reduced financial risk: If the government provides coverage for chronic illnesses, private insurance companies may experience reduced financial risk associated with high claims costs for chronic conditions.
- 2) Increased competitiveness: Insurance companies may face less pressure to cover chronic illnesses comprehensively, allowing them to focus on other areas where they can offer unique value propositions and compete effectively.
- 3) Potential for collaboration: Insurance companies could potentially collaborate with the government to offer supplementary coverage or specialized services for chronic illnesses, leveraging their expertise and resources.

Disadvantages:

- 1) Loss of market share: Government-provided coverage for chronic illnesses may lead to decreased demand for private insurance products covering the same conditions, resulting in loss of market share for insurance companies.
- 2) Regulatory challenges: Insurance companies may face additional regulatory burdens or constraints if the government becomes directly involved in providing coverage for chronic illnesses, potentially limiting their flexibility and innovation.

- 3) Profitability concerns: Government involvement in providing coverage for chronic illnesses could impact the profitability of private insurance companies, particularly if they are unable to compete effectively or maintain sufficient premiums to cover costs.

II. Government

Advantages:

- 1) Improved access to healthcare: Government-provided coverage for chronic illnesses can help ensure that all citizens have access to essential healthcare services and treatments, regardless of their financial status or insurance coverage.
- 2) Cost containment: By covering chronic illnesses, the government may be able to leverage economies of scale and negotiate lower prices for healthcare services and medications, helping to contain overall healthcare costs.
- 3) Public health benefits: Government involvement in providing coverage for chronic illnesses can contribute to improved public health outcomes, reduced healthcare disparities, and enhanced quality of life for affected individuals.

Disadvantages:

- 1) Budgetary constraints: Providing coverage for chronic illnesses can impose significant financial burdens on government budgets, especially if the prevalence of chronic conditions is high or if costs exceed initial projections.
- 2) Administrative challenges: Managing and administering coverage for chronic illnesses can be complex and resource-intensive for government agencies, requiring robust infrastructure, personnel, and oversight mechanisms.
- 3) Potential for political scrutiny: Government involvement in healthcare provision, including coverage for chronic illnesses, is often subject to political scrutiny and debate, leading to challenges in policy implementation and continuity.

III. Customers

Advantages:

- 1) Expanded coverage options: Government-provided coverage for chronic illnesses can supplement existing private insurance policies, providing additional options for policyholders to access necessary healthcare services and treatments.
- 2) Financial relief: Policyholders may experience financial relief if the government covers certain chronic illnesses, reducing out-of-pocket expenses for medical care and medications.
- 3) Enhanced peace of mind: Knowing that chronic illnesses are covered by either private insurance or the government can provide peace of mind to policyholders and alleviate concerns about financial insecurity in the event of serious health issues.

Disadvantages:

- 1) Limited choice and flexibility: Government-provided coverage for chronic illnesses may limit policyholders' choices in terms of healthcare providers, treatments, and coverage options compared to private insurance plans.

- 2) Quality of care concerns: Policyholders may have concerns about the quality and accessibility of healthcare services under government-provided coverage for chronic illnesses, particularly if there are shortages of healthcare providers or long wait times for appointments and treatments.
- 3) Potential for coverage gaps: There may be gaps in coverage between private insurance plans and government-provided coverage for chronic illnesses, leading to confusion or gaps in healthcare services for policyholders.

In summary, while government involvement in providing coverage for chronic illnesses can offer advantages such as improved access to healthcare and cost containment, it also presents challenges related to budgetary constraints, administrative complexities, and potential impacts on private insurance companies and policyholders. Careful consideration of these factors is essential for policymakers when evaluating the merits of government-provided coverage for chronic illnesses.

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Solution 2:

a) Typical features of a PMI product:

- 1) indemnity product that seeks to provide compensation for the cost of private medical treatment
- 2) provides cover for acute rather than chronic conditions
- 3) usually, the insurer pays the treatment provider directly
- 4) yearly renewable – premium can vary from year to year
- 5) requires minimum period of hospitalization
- 6) coverage of day care treatment
- 7) mostly in-patient only but may cover out-patient treatment to some extent
- 8) some form of co-insurance or sub-limits to prevent overuse
- 9) Excludes Pre-existing conditions, GP services (i.e. visits to your doctor) and cosmetic treatments
- 10) Many policies now cover types of treatment such as AYUSH, homeopathy and acupuncture
- 11) General exclusions: Drug abuse, self-inflicted injuries, hazardous pursuits etc.

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b) Uncertainty around cost of cover for PMI in a new market using another market's data:

- 1) Differences in lifestyle and prevalence of medical conditions
 - e.g. obesity linked to heart disease and diabetes
 - Leads to differences in frequency of hospitalization claims
- 2) Difference in demographics
 - Aging population, leads to increased demand for healthcare services
- 3) Effect of innovation on the cost of medical treatment covered
 - e.g. Availability/use of more sophisticated medical technology may increase the cost of cover
 - e.g. Innovations in surgical procedures may shorten the required recuperation time in hospital
 - e.g. Availability of health care resources – increased access increases utilisation, and vice versa
 - e.g. Differences in availability of clinics or cheaper facilities for less serious conditions

- Leads to differences in cost of care
- 4) Differences in policyholder and healthcare provider behaviour
 - e.g. could result in higher utilisation of medical services
- 5) Differences in claims control and frauds
 - Poor claims control can result in insurer paying claims that should have been excluded due to benefit limitations
- 6) Effect of introducing claims management processes such as preferred providers or managed care
 - e.g. negotiated fees
- 7) Currency fluctuations
 - Less developed country vs well developed country and the currency rates with exporting countries
 - leads to volatility in cost of medical treatment that relies on imported equipment or pharmaceuticals
- 8) Regulation overseeing products offered
 - Differences in cover, benefit limitations and exclusions
- 9) Differences in distribution channels
 - Different mix by distribution channel may also result in different claiming patterns and experience
- 10) Difference in Taxation and Legal environment and likely changes in the future
- 11) Political stability and Government approach towards health services in a less developed market
- 12) Differences in social set-up
 - e.g nuclear family vs joint family

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- c) Electronic health records help to store a wide range of data points related to individuals' health and healthcare utilization.
1. Basic Demographic Information: This includes details such as name, address, date of birth, gender, and contact information.
 2. Insurance Coverage Information: Details about the individual's insurance plan(s), including policy number, coverage start and end dates, and type of coverage (e.g., individual, family, employer-sponsored).
 3. Medical History: Information about past illnesses, surgeries, medical procedures, allergies, and chronic conditions.
 4. Medication History: A record of prescribed medications, dosage, frequency, and duration of use.
 5. Immunization Records: Details about vaccinations received, including dates and types.
 6. Lab Results: Test results from blood work, urine analysis, imaging tests, and other diagnostic procedures.
 7. Diagnostic Reports: Reports from medical imaging (e.g., X-rays, MRIs, CT scans), pathology reports, and other diagnostic tests.
 8. Treatment History: Information about treatments received, including dates, types of treatment (e.g., surgeries, therapies), and outcomes.

9. Encounter Information: Records of visits to healthcare providers, including dates, providers seen, reason for visit, and services provided.
10. Care Plans: Documentation of care plans developed in collaboration with healthcare providers, including goals, interventions, and outcomes.
11. Health Risk Assessments: Information collected through assessments to identify health risks and develop appropriate interventions.
12. Health Behavior Information: Data on lifestyle factors such as diet, exercise, smoking status, alcohol consumption, and substance use.
13. Family Medical History: Details about the health conditions and medical history of the individual's family members.
14. Provider Information: Details about healthcare providers seen by the individual, including names, specialties, contact information, and affiliations.
15. Billing and Claims Data: Information related to healthcare services rendered, including billing codes, claim submissions, payments, and reimbursements.
16. Consent and Authorization Forms: Records of individuals' consent for sharing their health information and authorizations for specific uses or disclosures.
17. Emergency Contact Information: Contact details for individuals designated as emergency contacts.

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d) Advantages and disadvantages of using the electronic record:

Advantages

- 1) Marketing:
 - a. Positively received by policyholders, leading to increased volumes of business or promoting existing policyholders to stay with the insurer
 - b. Positively received by providers, assisting with building relationships
 - c. Competitive advantage if tool is perceived to be useful and/or the best Better data
- 2) More complete data set for the insurer, resulting in more accurate pricing
- 3) Insurer receives data timeously:
 - a. reducing lag to receive data
 - b. Helping to refine IBNR reserve
- 4) Improve the care provided by doctor as they have all the information required to make informed decisions
- 5) Less duplication of tests saving on the costs of these tests
- 6) Reduce risk of error in data due to manual capturing, improving the quality of the data
- 7) Can have checks in place to capture data correctly, e.g. alphanumeric vs numeric vs special characters
- 8) Can encourage or enforce certain data fields to be input, e.g. ICD10 codes
- 9) Collect clinical results, resulting in better understanding of the epidemiology of the policyholders (leading to more accurate pricing) and better understanding of the quality of care provided
- 10) Large amounts of data to track different types of health hazards linked either to medical, environmental or social factors Submitting claims
- 11) Faster and easier

- 12) Can use the tool to apply for pre-auth and grant/deny it
- 13) Can determine co-payments applicable or benefit limits reached/available etc. via the tool immediately Save on costs
- 14) Less duplication of tests, and less unnecessary tests
- 15) Preventing hospitalisation if appropriate treatment protocols are followed correctly
- 16) Efficiencies gained by electronic transfer of data instead of manual data capturing
- 17) Improved coordination of care improves health status

Reduce admin burden

- 1) Assist doctors with creating and managing appointments
- 2) Real-time alert to doctors on patients' condition, tests required, time elapsed since last consultation etc. Open communication with doctors
- 3) Educate doctors on best practices
- 4) Give guidance on treatment protocols
- 5) Alerts doctors to potential treatment complications such as co-morbidities
 - a. improving quality of care.
- 6) Educate on latest research findings and generics released
- 7) Provides support to doctors for decisions made in care Disadvantages Poor uptake
- 8) Will render all advantages meaningless Legal considerations/ Regulation risk
- 9) Privacy issues around what information can and cannot be shared
- 10) Risk of providing incorrect data - Who does that data belong to (policyholder, provider, insurer)
- 11) Needs adequate control and standards
- 12) Regulators may have minimum requirements before it can be launched, including ongoing reports to demonstrate compliance

IT security / Data security

- 1) risk of being hacked and data stolen. Increase costs
- 2) Development expenses, including possibly needing more IT staff, more space and additional systems
- 3) May increase cost of cover by doing more screening and more tests
- 4) Training sessions to train doctors to understand and use the tool, and to train insurer's staff to field questions about it and to use the data, and train brokers to explain tool to policyholders

Reputational risk

- 1) Confusing to doctors and policyholders - Risk that the tool doesn't work well
- 2) Added admin burden to doctors, including data overload
- 3) Patients may get less dedicated time with doctor in the room if the doctor is focused on the electronic record and all the fields needed to be filled in Competition
- 4) How does the tool compare to the tool offered by competitors
- 5) Will doctors be able to learn to use both tools Systems integration
- 6) Will the tool work with the systems that the providers have available, across all providers in the care chain

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e) Corrective actions

For appropriate action to be taken, it is imperative that the company understands the reasons for any adverse trends in the experience.

- 1) Update assumptions for future experience
 - e.g. understand if trends are changing and update assumptions so that they are appropriate for expected future experience - Re-pricing of products
 - e.g. include larger margins subject to competitive pricing
- 2) Re-design of products
 - e.g. through introducing benefit limits or through stricter referral patterns or protocols in order to access benefits –
- 3) Change the investment strategy

- e.g. an investment strategy that maximises investment returns for the acceptable level of risk
- 4) Change the sales strategy
 - e.g. which products are marketed through which distribution channel, change the commission structures etc., target populations with the best claims experience/the most profitable group
 - 5) Change the reinsurance strategy
 - e.g. takeout reinsurance to protect against future adverse experience
 - 6) Change the underwriting strategy
 - e.g. apply stricter underwriting in order to better price poor risks or to exclude members that do not meet the eligibility criteria
 - 7) Change the profit distribution strategy
 - e.g. to shareholders
 - 8) Re-organise the workforce and IT systems to make more efficient use of expensive resources
 - 9) Change managed care arrangements
 - e.g. negotiated fees, introduce risk sharing agreements

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